

JET IMPACT ON SURFACES

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1 Goal of the lab session

The objective of this lab is to experimentally measure the force exerted by a water jet on three surfaces with different geometries and to compare the experimental results obtained with those predicted by the theoretical analysis of the problem.

2 Problem definition and theoretical framework

Consider the situation shown in Figure 1, where a water jet impinges onto a solid surface: (a) flat, (b) inclined at 30° , or (c) hemispherical. The jet carries a water flow rate of Q and exits from a nozzle with an inner diameter of $d = 8$ mm. Therefore, the average velocity of the jet at the nozzle exit is $v = Q/A$, where $A = \pi d^2/4$ is the cross-sectional area of the jet.

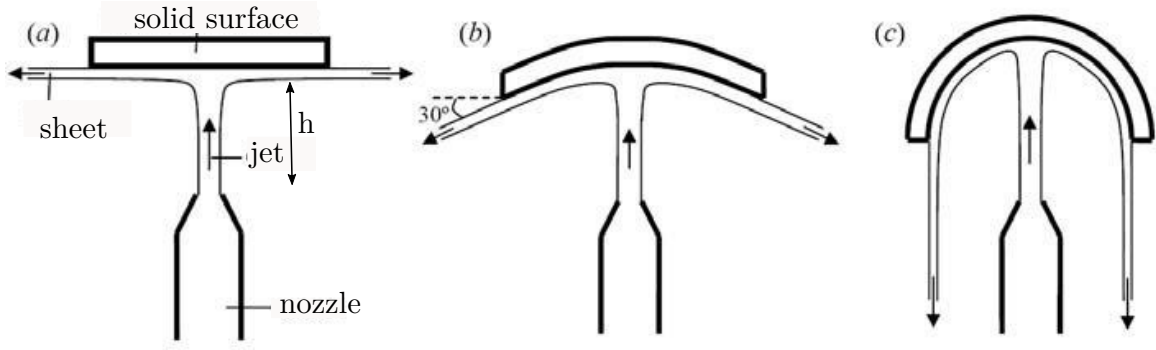


Figure 1: Geometric configurations of jet impact considered. Impact on a (a) flat surface, (b) inclined surface, and (c) hemispherical surface.

After impacting the surface, the water flow is ejected with a mean velocity v_s , tangential to the surface, forming a round water sheet with a cross-sectional area A_s . Both viscous and gravity effects can be neglected, since the characteristic Reynolds number, defined as $Re_d = \rho v d / \mu \gg 1$ where ρ and μ are the density and viscosity of water, and Froude number, defined as $Fr = v / \sqrt{g h} \gg 1$ where g is the gravitational acceleration and h is the distance between the nozzle exit and the solid surface, are large. Therefore, under steady-state conditions ($Q = \text{constant}$) it is possible to apply Bernoulli's equation, which along the free surface of the jet results in $v_s = v$, meaning that the velocity of the fluid after impacting the surface is equal to the jet velocity at the nozzle exit. The equivalence of velocities, together with mass conservation, implies that the cross-sectional area of the sheet must equal the cross-sectional area of the jet, $A_s = A$. Finally, by applying the momentum conservation equation, the total force on the plate can be obtained (left as an exercise for the student) as a function of the exit angle of the water sheet. Thus, the force of the jet on the plate in each case is

$$(a) F = \rho Q^2 / A, \quad (b) F = 1.5 \rho Q^2 / A, \quad \text{y} \quad (c) F = 2 \rho Q^2 / A. \quad (1)$$

If we define the drag coefficient as the dimensionless number

$$C_d = \frac{F}{\frac{1}{2} \rho v^2 A}, \quad (2)$$

it is easy to verify that it takes the values $C_d = 2$ for the flat surface, $C_d = 3$ for the case of the oblique surface, and $C_d = 4$ for the hemispherical surface.

3 Description of the Experimental Equipment

In the experiment, the three situations shown in Figure 1 will be considered: impact on a flat surface (a), an oblique surface (b), and a hemispherical surface (c). To perform the measurements, the hydraulic bench shown in Figure 2 will be used, which is capable of supplying a constant flow rate and has the necessary elements for measuring that flow rate.

Figure 3 shows the measuring device that is mounted on top of the bench, which will allow us to measure the force exerted by the water jet by balancing this force with the addition of calibrated weights.



Figure 2: Image of the hydraulic bench.



Figure 3: Experimental force-measuring device.

4 Measurement Procedure

The task consists of measuring the force exerted by the jet on a surface for each of the three selected configurations. To that end, the following experimental procedure must be followed:

- Remove the cover resting on the transparent tank of the reservoir. Place the flat surface in the impact position fixed to the vertical bar that is part of the force calibration system. Close the tank again.
- Place the assembly in the channel of the hydraulic bench, connecting its water inlet to the pump.
- Level the assembly.
- Adjust the dial gauge until it is at the same height as the signal from the auxiliary platform.
- Place a mass on the platform. Start the pump of the hydraulic bench and allow water to flow, regulating the flow rate until the signal from the platform is at the same height as the reading of the dial gauge.
- Once equilibrium is reached, record the values of the outflow rate from the nozzle and the mass on the platform.
- Repeat the previous operations, incrementally increasing the mass on the surface.
- Repeat the entire process for the inclined and hemispherical surfaces.

5 Experimental Results

The experimental measurements of mass and flow rate, as well as the calculations of weight and force, should be recorded as indicated in Table 1.

Contents of the Laboratory Report

A report will be submitted that includes:

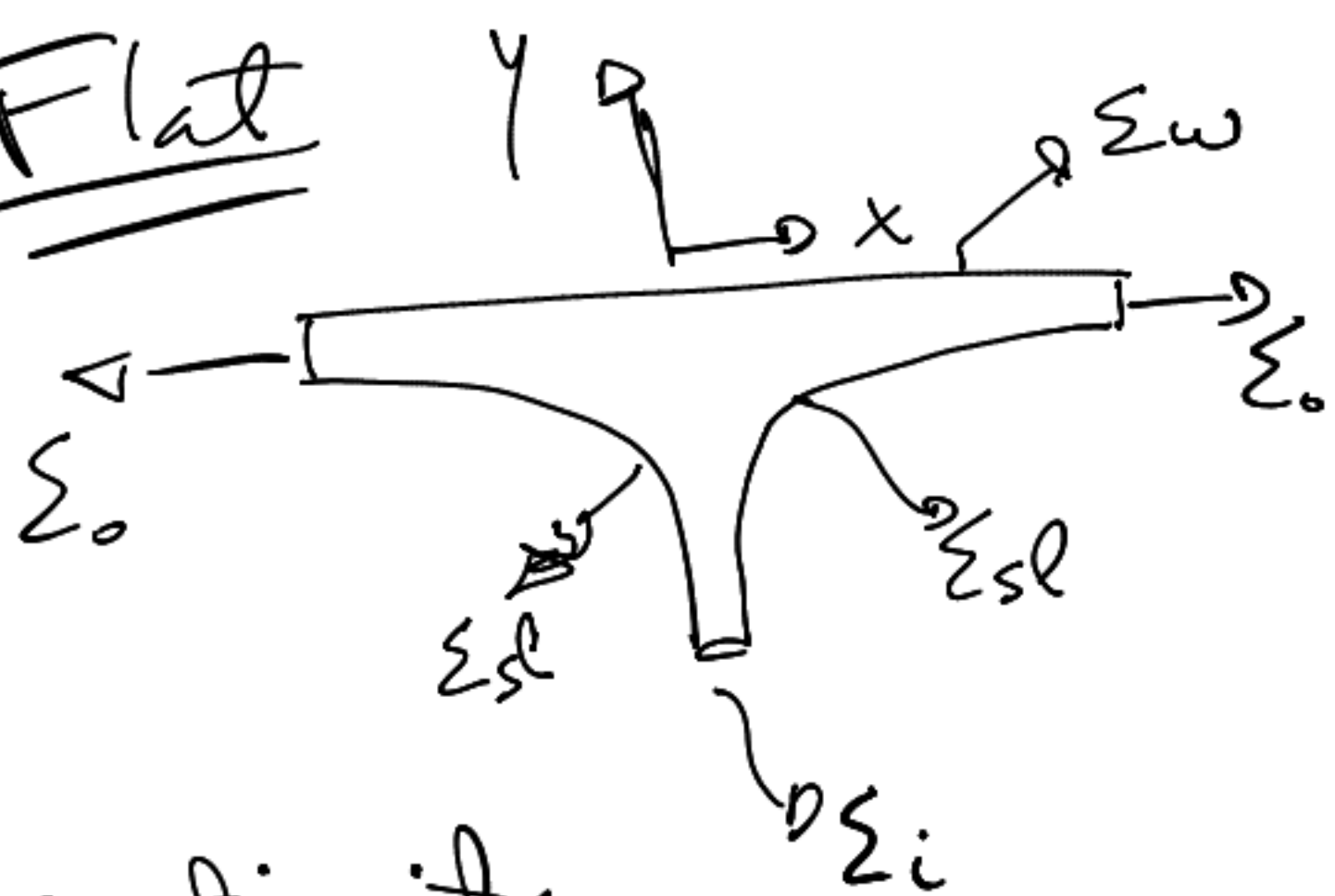
- A brief introduction.
- Demonstration of the theoretical expressions (Equation 1) for the force exerted by the jet on the three geometries. To do this, it is suggested to use the continuity equation and the momentum equation projected in the direction of the jet, applied to a control volume bounded by the jet section at the nozzle exit, the cross-section of the water sheet leaving the plate, the free surface of the jet, and the wetted wall of the obstacle. Note that the atmospheric pressure acting on the dry face of the obstacle must be included to obtain the total force on the plate.
- Description of the experimental method.
- Tables of experimental data.
- For each obstacle, a graph of the experimental data of F versus Q , represented with unconnected points, along with the corresponding theoretical prediction as a continuous line.
- In a single figure, comparison of the experimental and theoretical data of F as a function of Q for the three tested surfaces.
- Calculation of C_d for each obstacle.
- Graphically represent the values of C_d against Re_d obtained experimentally for the three obstacles.
- Comments and conclusions.

Flat surface				
	m (g)	Q (l/h)	W (N)	$\frac{1}{2}\rho Q^2/A$ (N)
1	50			
2	100			
3	150			
4	200			
5	...			
6				
7				
8				
9				
10				

Oblique surface				
	m (g)	Q (l/h)	W (N)	$\frac{1}{2}\rho Q^2/A$ (N)
1	50			
2	100			
3	150			
4	200			
5	...			
6				
7				
8				
9				
10				

Hemispherical surface				
	m (g)	Q (l/h)	W (N)	$\frac{1}{2}\rho Q^2/A$ (N)
1	50			
2	100			
3	150			
4	200			
5	...			
6				
7				
8				
9				
10				

Flat



Bernoulli

$$p_a + \frac{1}{2} \rho v_i^2 + \rho g z_A = p_e + \frac{1}{2} \rho v_0^2 + \rho g z_B$$

$$v_i = v_0$$

Continuity

$$\int_{\Sigma_c} \rho \vec{v} \cdot \vec{n} d\epsilon = \int_{\Sigma_i} \rho v_i \vec{e}_y \cdot (-\vec{e}_y) d\epsilon + \int_{\Sigma_{se} + \Sigma_w} \rho \vec{v} \cdot \vec{n} d\epsilon + \int_{\Sigma_o} \rho \vec{v} \cdot \vec{n} d\epsilon = 0$$

$$-\rho v_i A_i + \int_{\Sigma_o} \rho v_0 \vec{e}_x \cdot \vec{e}_x d\epsilon = 0$$

$$\rho v_i A_i = \rho v_0 A_o \rightarrow A_i = A_o$$

Momentum

$$\int_{\Sigma_c} \rho \vec{v} (\vec{v} \cdot \vec{n}) d\epsilon = - \int (\rho - \rho_a) \vec{n} d\epsilon$$

$$\Sigma_c = \Sigma_i + \Sigma_o + \Sigma_{se} + \Sigma_w$$

$$\int_{\Sigma_i} \rho v_i \vec{e}_y (-v_i) d\epsilon + \rho v_0^2 A_o \vec{e}_x = -\vec{F}$$

$$\vec{F} = \rho v_i^2 A_i \vec{e}_y - \rho v_0^2 A_o \vec{e}_x$$

$$F_v = \rho v_i^2 A_i = \rho v_i^2 A_i \frac{A_i^2}{A_i^2} = \rho \frac{Q^2}{A}$$

Hemispherical

Momentum

$$\int \rho \bar{v} (\bar{v} \cdot \bar{n}) d\phi = - \int (p - p_a) \bar{n} d\phi$$
$$\Sigma_c = \Sigma_i + \Sigma_o + \cancel{\Sigma_{se}} + \cancel{\Sigma_w} \quad \Sigma_c = \cancel{\Sigma_i} + \cancel{\Sigma_o} + \cancel{\Sigma_{se}} + \Sigma_w = -\bar{F}$$

$$\int_{\Sigma_i} \rho v_i \bar{e}_y (-v_i) d\phi + \int_{\Sigma_o} \rho (-v_o \bar{e}_y) (-v_o \bar{e}_y \cdot (-\bar{e}_y)) d\phi = -\bar{F}$$

$$-\rho v_i^2 A_i \bar{e}_y - \rho v_o^2 A_o \bar{e}_y = -\bar{F} \quad \begin{matrix} v_i = v_o = v \\ A_i = A_o = A \end{matrix}$$

$$\bar{F} = 2\rho v^2 A \bar{e}_y$$

$$F_v = 2\rho v^2 A = 2\rho \frac{Q^2}{A}$$

Oblique

$$-\rho v_i^2 A_i \bar{e}_y + \int_{\Sigma_o} \rho (-v_o \cos(\pi/6) \bar{e}_x - v_o \sin(\pi/6) \bar{e}_y) (\underbrace{\bar{v}_o \cdot \bar{n}_o}_{v_o \cdot \frac{1}{2}}) d\phi = -\bar{F}$$

$$\bar{F} = \rho v_i^2 A_i \bar{e}_y + \rho v_o^2 \cos(\pi/6) A \bar{e}_x + \rho v_o^2 \sin(\pi/6) A \bar{e}_y$$

$$F_v = (1 + \frac{1}{2}) \rho v^2 A = \frac{3}{2} \rho \frac{Q^2}{A}$$